

★ BEST PERFORMANCE IN ★ SCIENCE OLYMPIADS

A LEGACY OF EXCELLENCE. A FUTURE OF IMPACT.

	3 GOLD		IPhO International Physics Olympiad
	2 GOLD		IBO International Biology Olympiad
	1 GOLD		IOAA International Olympiad on Astronomy & Astrophysics
	1 GOLD		IOAA (Jr) International Olympiad on Astronomy & Astrophysics Jr.
	1 GOLD		APhO Asian Physics Olympiad



	2 SILVER		IOAA International Olympiad on Astronomy & Astrophysics
	1 SILVER		APhO Asian Physics Olympiad
	1 BRONZE		IOAA (Jr) International Olympiad on Astronomy & Astrophysics Jr.
	1 GOLD		IJSO International Junior Science Olympiad
	1 SILVER		IJSO International Junior Science Olympiad
	2 SILVER		ICHO International Chemistry Olympiad



FOCUS



PREPARE



ACHIEVE



INSPIRE



2020 - 2026

DEDICATION. DISCIPLINE. EXCELLENCE.
ONE JOURNEY. ONE TEAM. ONE LEGACY.

Vedantu

Master Every Concept



**DREAM. PREPARE.
ACHIEVE. REPEAT.**

Nidhi Prashar

Chemistry



- **16+ Years of teaching experience**
- **Gold Medallist in the Master's Degree (Chemistry), UGC NET Qualified**
- **IJSO EXPERT with 170+ NSEJS Selections**
- **Mentored Science Olympiads GOLD MEDALISTS**
- **Teaching and mentoring students IIT, NEET, NSEC, IJSO exams**
- **Produced Single and double digit ranks in JEE/NEET (AIR 4 in AIIMS)**

ABHIMANYU BATCH-2026 SELECTION TEST *IS HERE!*



Choose your slot & secure your chance
to join the most awaited batch of the year



SELECTION TEST 1
16TH AUGUST



7:00 PM
ONWARDS



SELECTION TEST 2
23RD AUGUST



11:00 AM
ONWARDS

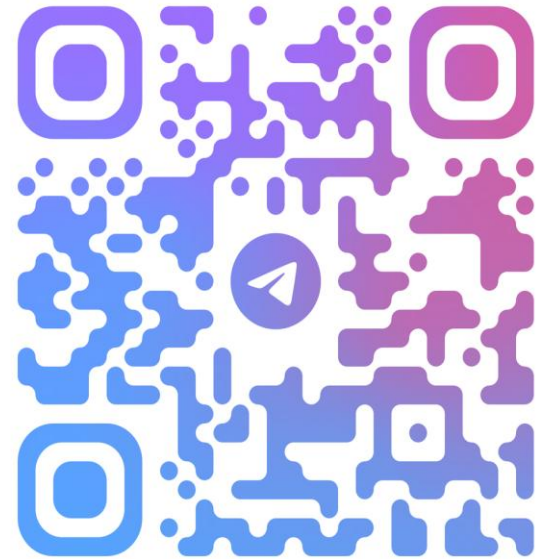


JOIN THE **VEDANTU SCIENCE OLYMPIAD**
TELEGRAM CHANNEL
TO GET THE TEST LINKS, UPDATES,
AND IMPORTANT INSTRUCTIONS.



TELEGRAM LINK IS PINNED IN THE COMMENTS

DON'T MISS THIS OPPORTUNITY TO BE A PART OF
ABHIMANYU BATCH. SEE YOU IN THE TEST! 



@VEDANTU_SCIENCE_OLYMPI
AD

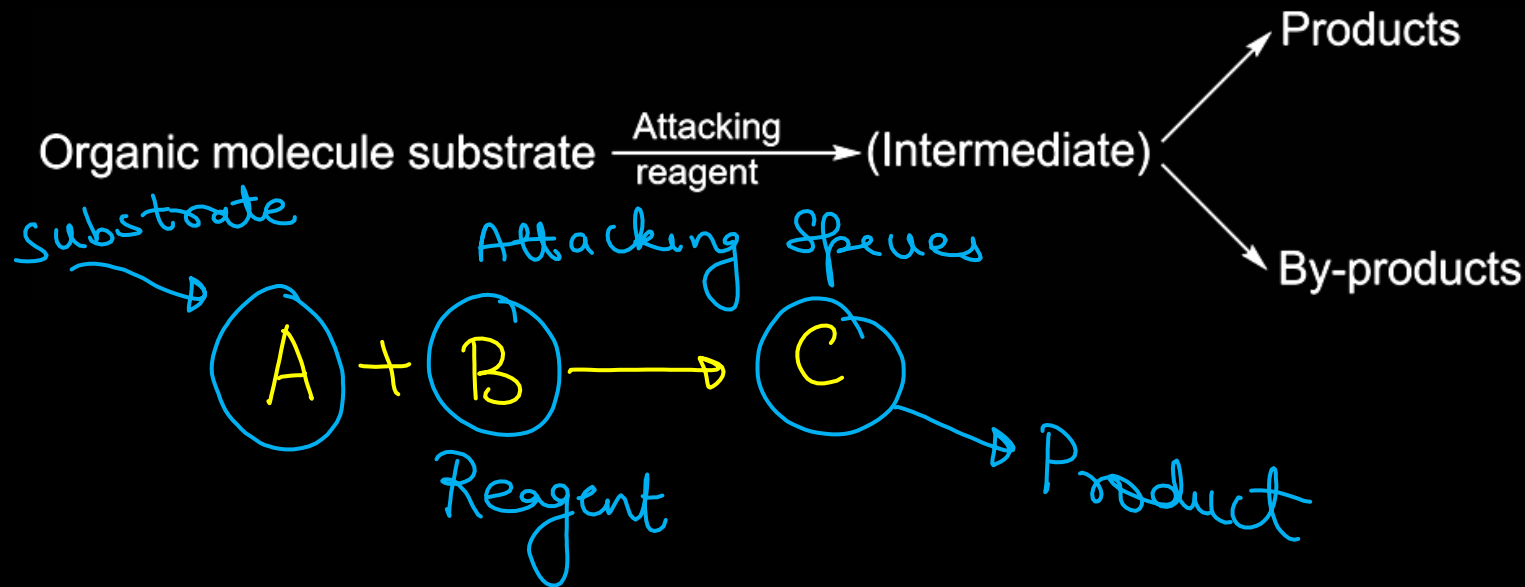
→ Basics
→ Nomenclature
→ Isomerism

} One-Shot

Mechanism → FIR Most Imp
↳ Inductive effect
↳ Resonance

FUNDAMENTAL CONCEPTS IN ORGANIC REACTION MECHANISM

The organic molecule (also referred as a substrate) reacts with an appropriate attacking reagent and leads to the formation of one or more intermediate(s) and finally product(s).



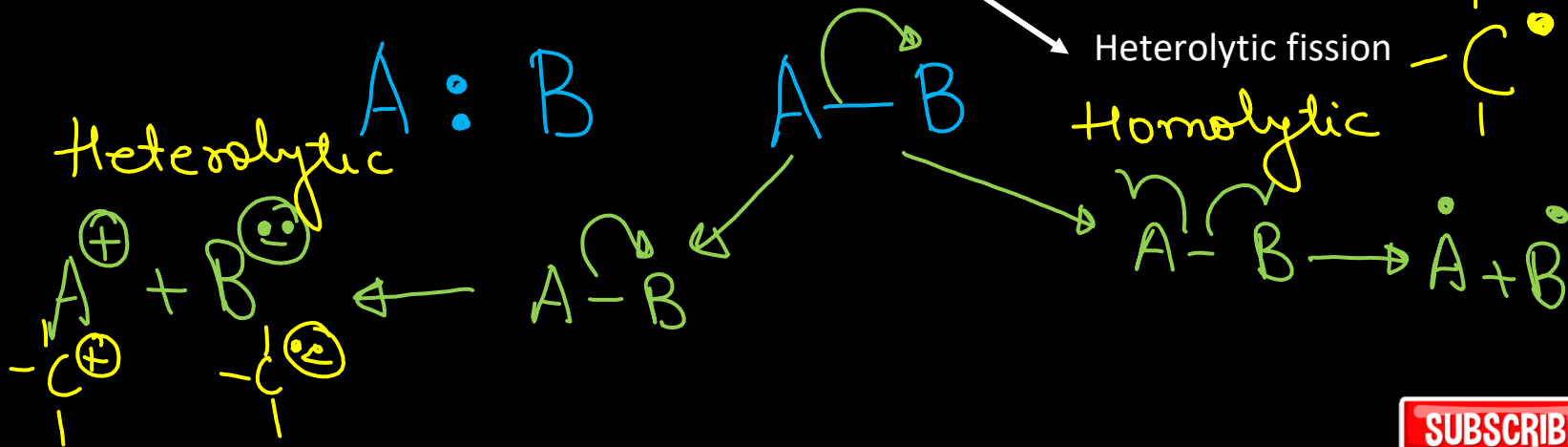
FUNDAMENTAL CONCEPTS IN ORGANIC REACTION MECHANISM

Reaction is **breaking** of old bonds and formation of new bonds

BOND FISSION/RUPTURE/CLEAVAGE/LYSIS

Homolytic fission

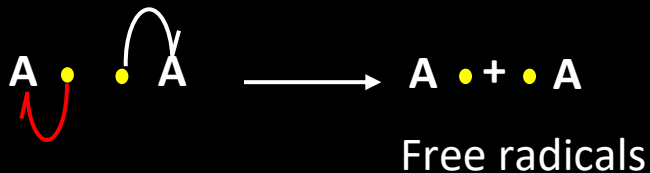
Heterolytic fission



SUBSCRIBE

Homolytic fission or Symmetrical Fission :

Representation : Half headed /Fish hooked arrow



 single e^- transfer

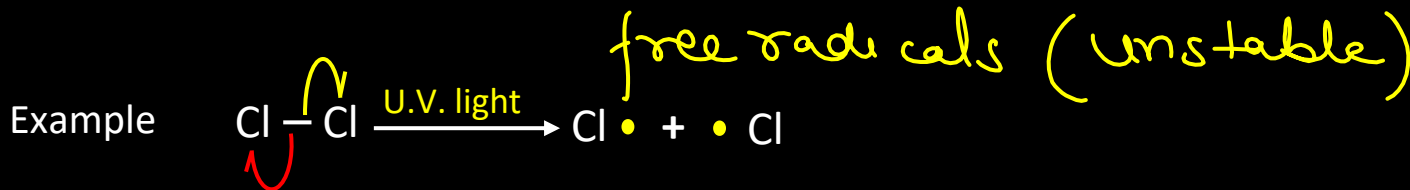
Homolytic fission or Symmetrical Fission :

Characteristics :

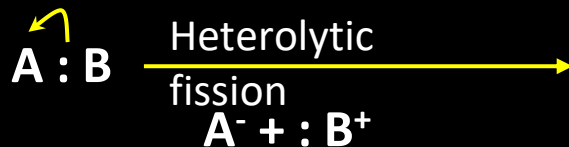
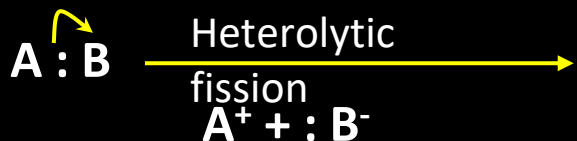
- 👉 Symmetrical breaking of bond
- 👉 Results in the formation of free radicals
- 👉 Usually takes place between two atoms having no difference OR small difference in the EN value
- 👉 Favoured by presence of HELPR

Symmetrical breaking of a covalent bond such that both participating atom's retain one e^- each resulting in the formation of free radicals

H = Heat, E = Electricity, L = Light, P = Peroxide, R = Radical



Heterolytic fission or Unsymmetrical fission :



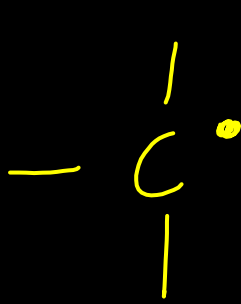
when A is more electronegative than B)

An unsymmetrical breaking of a covalent bond such that any one of the two participating atom's retain bonding e^- pair.

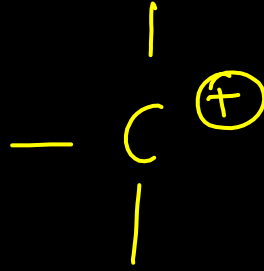
Heterolytic fission or Unsymmetrical fission :

Characteristics :

- 👉 Unsymmetrical breaking of a bond.
- 👉 Results in the formation of free ions.
- 👉 Usually takes place between the atoms differing in their EN values.
- 👉 Favored by presence of polar solvents.



Carbon
free
radical



Carbocation



Carbanion

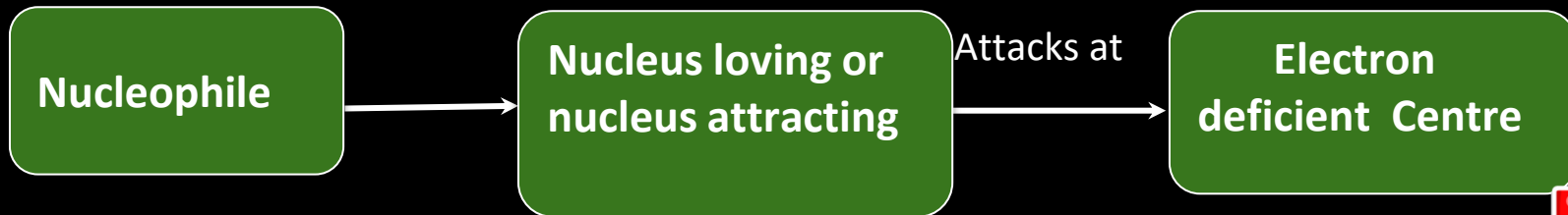
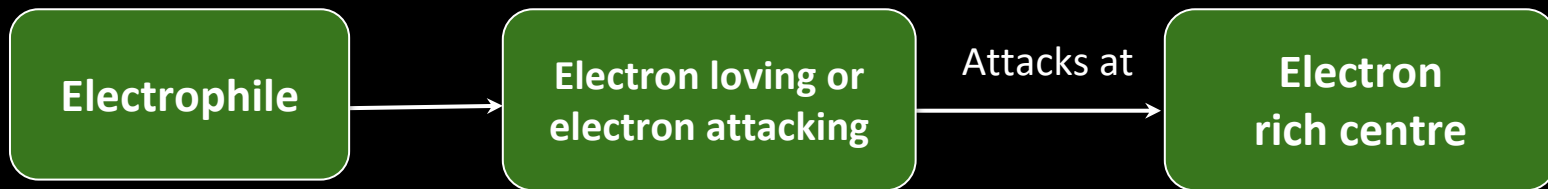
Reaction Intermediates

Types Of Attacking Reagents :

The attacking reagents are classified as

👉 Electrophiles $\rightarrow e^-$ loving

👉 Nucleophiles $\rightarrow$ nucleus loving



SUBSCRIBE

Electrophiles :

A reagent that takes away an electron pair is called electrophile. There are positively charged or neutral species which are deficient of electrons and can accept a pair of electrons.

❖ **Electrophile / Electrophilic reagent** : [Electro = Electron ; philic = Loving]

Thus, electrophile are e^{-} deficient species which has a tendency to attack at e^{-} rich centre

Types of Electrophile :

❑ Positively charged electrophile : NO_2^{+} , CH_3^{+} , Br^{+} , Cl^{+} etc.

❑ Neutral Electrophile : AlCl_3 , SO_3 , PCl_3 , etc. ' (vacant d-orbitals)

SUBSCRIBE

Nucleophiles :

A reagent that brings an electron pair is called nucleophile. A nucleophile is electron-rich and seeks electron-deficient sites i.e., nucleus-loving or nucleus-seeking (Nu).

Nucleophile / Nucleophilic reagent : [Nucleo = nucleus ; philic = Loving]

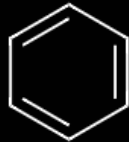


Nucleophiles :

Types of Nucleophile

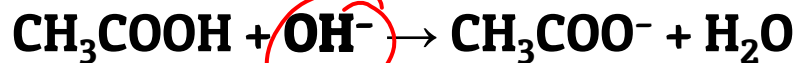
👉 Anionic Nucleophile : $\text{OH}^{\ominus}$, $\text{Cl}^{\ominus}$, $\text{CH}_3\text{O}^{\ominus}$, $\text{NH}_2^{\ominus}$ etc.

👉 Neutral Nucleophile : $\text{H}_2\ddot{\text{O}}:$, $\text{R}-\ddot{\text{O}}\text{H}$, $\text{R}-\ddot{\text{S}}\text{H}$, $\text{R}-\ddot{\text{O}}-\text{R}$ etc.

👉 π bonded Nucleophile :  , 

👉 Ambident Nucleophile : $:\text{C}\equiv\text{N}:$, $:\ddot{\text{O}}-\text{N}=\ddot{\text{O}}:$ etc.

Classify the reagents shown in bold in the following equations as nucleophiles or electrophiles.

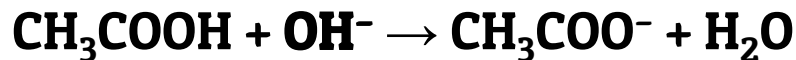


?

nucleophile

- A Nucleophile**
- B Electrophile**
- C Both possible**
- D Can't Predict**

Classify the reagents shown in bold in the following equations as nucleophiles or electrophiles.

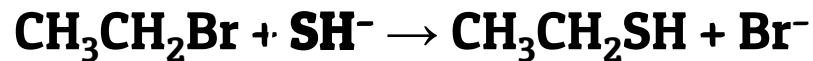


- A** Nucleophile
- B** Electrophile
- C** Both possible
- D** Can't Predict

Solution

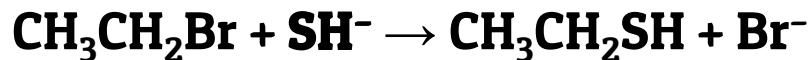
OH^- is a nucleophile.

Classify the reagents shown in bold in the following equations as nucleophiles or electrophiles.



- A Nucleophile
- B Electrophile
- C Both possible
- D Can't Predict

Classify the reagents shown in bold in the following equations as nucleophiles or electrophiles.

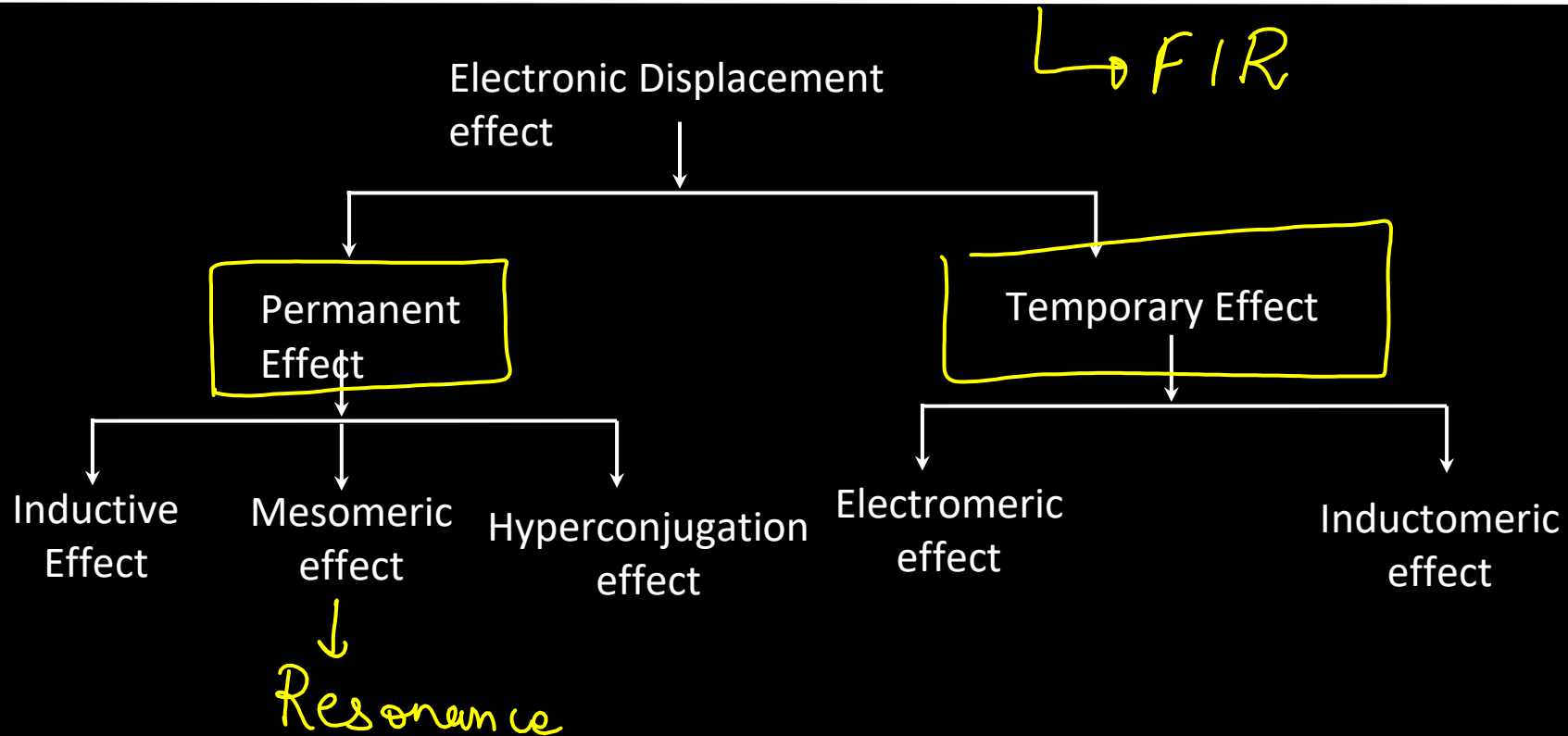


- A** Nucleophile
- B** Electrophile
- C** Both possible
- D** Can't Predict

Solution

SH^- is a nucleophile

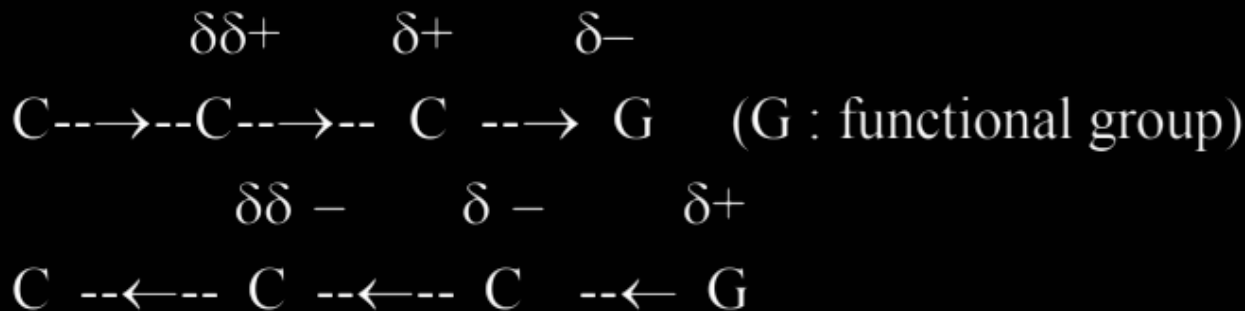
ELECTRONIC DISPLACEMENT EFFECTS

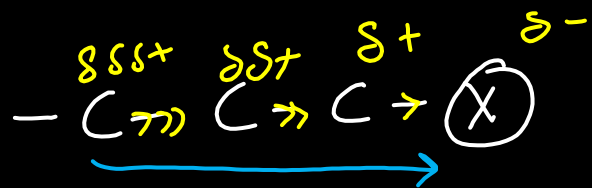


Inductive Effect

👉 Inductive Effect (I-effect):

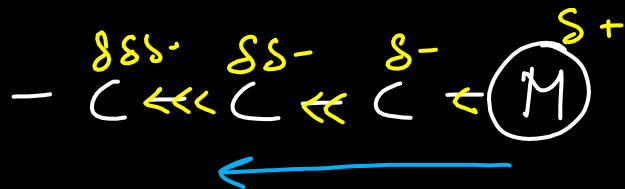
The displacement (shifting) of an electron (shared) pair along the carbon chain due to the difference in electronegativity of the groups present along with carbon chain is known as inductive effect.





ENG
-I-effect

X = Halogen
M = Metal



ERG
+I-effect



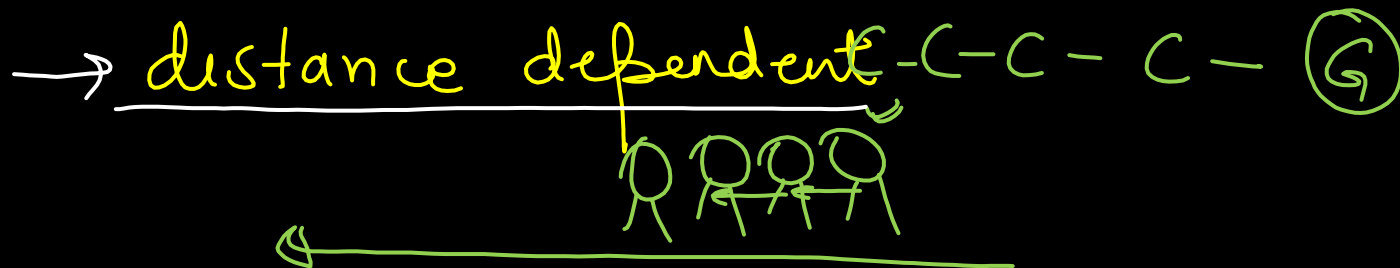
X

same
electronegative

Inductive effect :

Characteristics :

- Permanent effect. ✓ $\longrightarrow$ weak effect
- Takes place between the atoms bonded by a covalent bond.
- It is just a shifting of σ e^- pair and not a complete transfer. (σ -effect)
- It is observed due to difference in the EN values of the bonded atoms.



Inductive effect :

- 👉 I effect is transmitted through the chain of C-atom but its intensity decreases as the distance of withdrawing group from c – atom increases and practically becomes negligible after 3rd, 4th c - atom
- 👉 Since I effect is a displacement of σ bond (a strong bond) it's a weak effect and can be correlated to dipole moment (μ).

+I effect:

ERG / EDG

+I Effect : An atom or a group of atom's donates e^- s to other atom via σ bond are called as Electron donating groups (EDG) and an effect of donation of e^- s to other atom is called as +I effect.

👉 Some examples of EDG's in the decreasing order of donation ability are :

$-\bar{C}H_2 > \bar{N}H > -O^{\ominus} > -COO^- > \overset{3^\circ}{\text{tertiary alkyl}} > \overset{2^\circ}{\text{secondary alkyl}} > \overset{1^\circ}{\text{primary -alkyl}} > -CH_3 > -H$

alkyl $\rightarrow$ (ERG) $\rightarrow$ Bigger alkyl $\rightarrow$ more ERG

Diagram illustrating the structure of a branched alkyl group (tertiary alkyl) with carbon atoms connected by single bonds, showing the increasing electron-donating effect (ERG) as the alkyl group size increases.

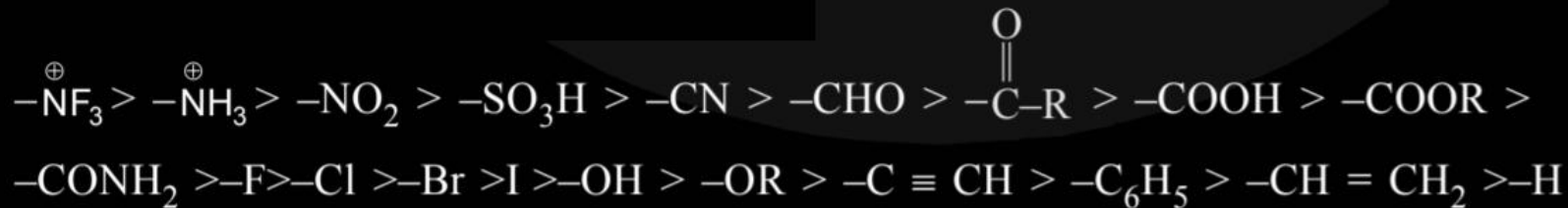
SUBSCRIBE

-I effect :

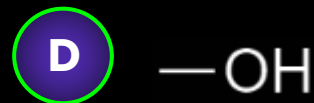
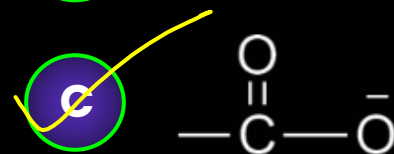
EWG

-I Effect An atom or a group of atom which withdraw electron from other atom through σ bond are called as Electron withdrawing (EWG) groups and the effect of withdrawal of e^- is called as -I effect

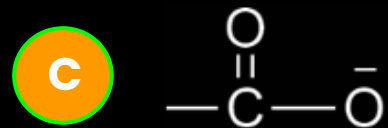
👉 Some examples of EWG in decreasing order of withdrawing ability are :



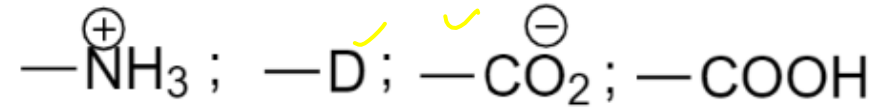
Which of the following belongs to +I group?



Which of the following belongs to +I group?



The Inductive effects of the groups



are respectively



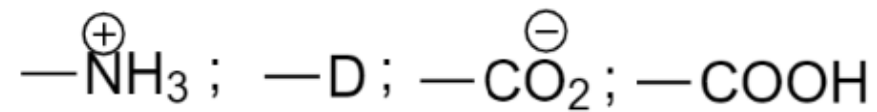
A $-I, \underline{+I, +I}, -I$

B $-I, -I, -I, +I$

C $+I, \text{No effect}, -I, -I$

D $+I, -I, -I, -I$

The Inductive effects of the groups



are respectively

A

-I, +I, +I, -I

B

-I, -I, -I, +I

C

+I, No effect, -I, -I

D

+I, -I, -I, -I

Strength of Carboxylic Acid

Acid strength is measured by the position of equilibrium of ionisation in water.



$$\text{or } K_a = \frac{[\text{H}^+][\text{A}^-]}{[\text{HA}]}$$

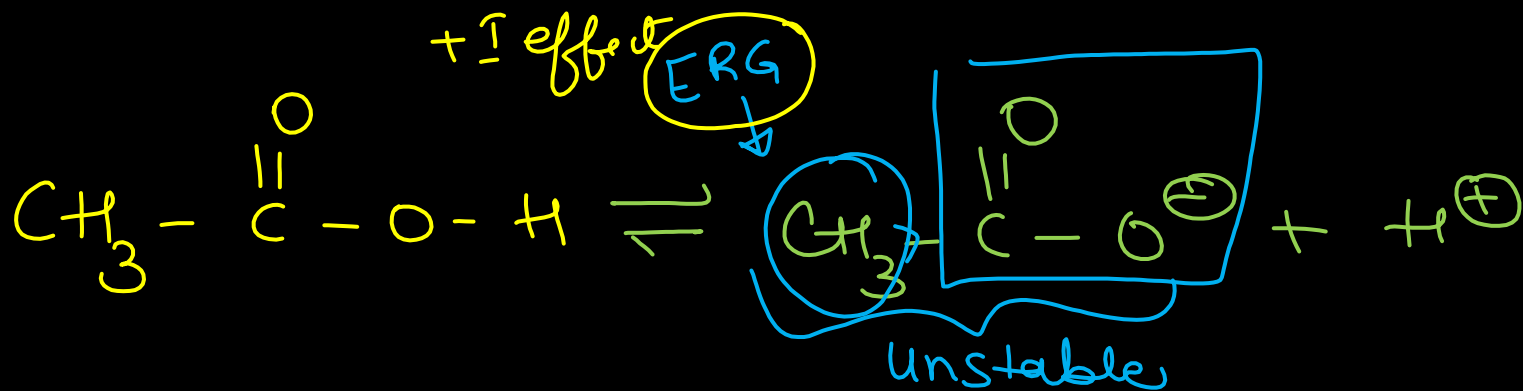
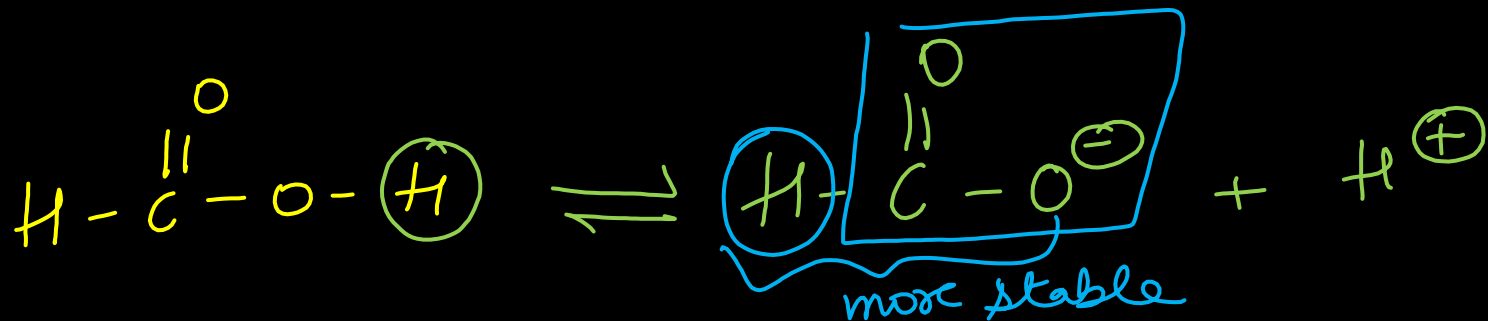
(K_a = acid ionisation constant)

$$\text{or } \text{acid strength} \propto [\text{H}^+] \propto \frac{1}{K_a}$$

(here $\text{p}K_a = -\log K_a$)

or Acid strength $\propto$ concentration of $[\text{H}^+]$

or Acid strength $\propto$ stability of acid anion

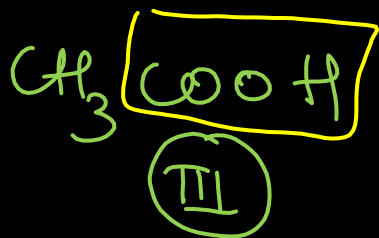
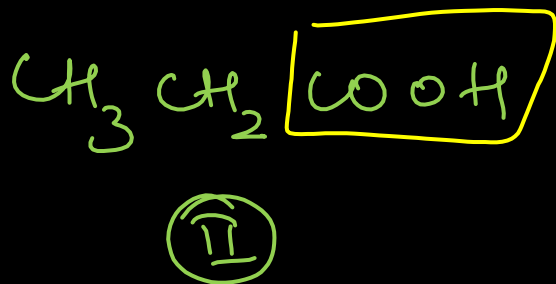
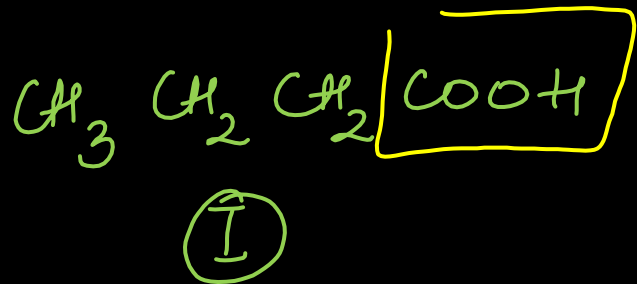


Acidic strength $\text{HCOOH} > \text{CH}_3\text{COOH}$

Note (v. Imp.) Remember

ERG (+I effect) $\Rightarrow$ decreases acidic strength

EWG (-I effect) $\Rightarrow$ increases acidic strength



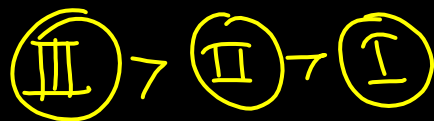
more acidic



K_a more



pK_a less



Acidic strength

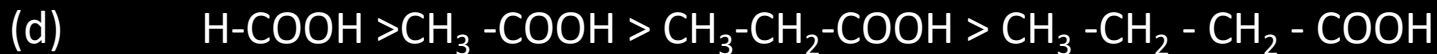
Strength of Carboxylic Acid

$\therefore \text{acidic strength} \propto \text{Presence of electron withdrawing group} \propto -I \text{ group} \propto \frac{1}{+I \text{ group}}$



$-I \uparrow$

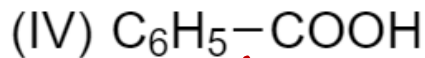
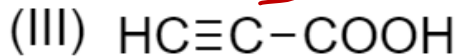
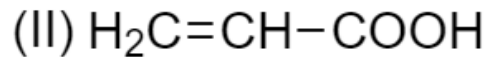
acid strength $\uparrow$



$+I$ power of alkyl groups $\uparrow$

Acid strength $\downarrow$

Find the order of acidic strength of following alcohols.



A

I > II > III > IV

B

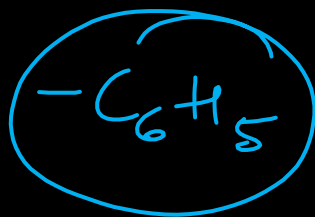
IV > III > II > I

C

I > III > II > IV

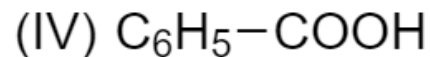
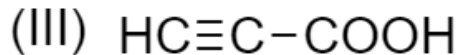
D

III > I > IV > II



- I effect

Find the order of acidic strength of following alcohols.



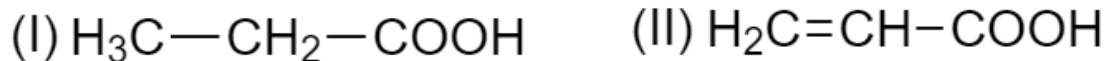
A I > II > III > IV

B IV > III > II > I

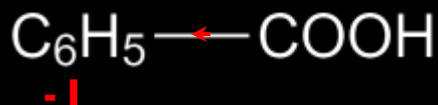
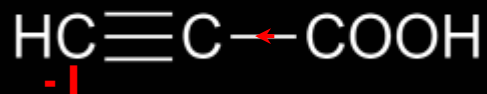
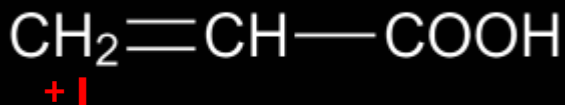
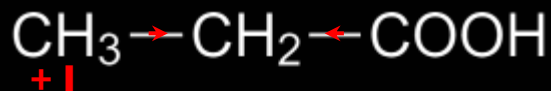
C I > III > II > IV

D III > I > IV > II

Find the order of acidic strength of following alcohols.



Solution :



Order of acidic strength : $\text{IV} > \text{III} > \text{II} > \text{I}$

Basicity of Amines



1° amine



2° amine



3° amine

(R) = alkyl

ERG

Gaseous/
non polar solvent

Basicity ↑ es

$3^\circ > 2^\circ > 1^\circ > \text{NH}_3$

Basicity of Amines

The groups producing +I effects (alkyl groups) tends to intensify electron density over N in amines thereby producing a base strengthening effect.

$$\text{Strength of base} \propto +I \text{ power of group present on } -\text{NH}_2 \propto K_b \propto \frac{1}{\text{pKb}}$$

Basicity of Amines

Whereas, a group producing **-I** effect [$-\text{Cl}$, $-\text{NO}_2$] tends to decrease electron density over N - atom in amines, thereby producing a base weakening effect.

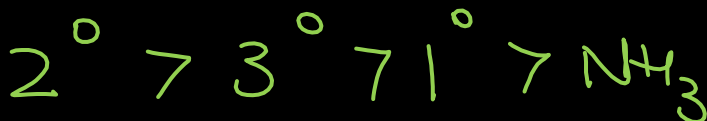
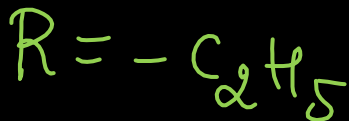
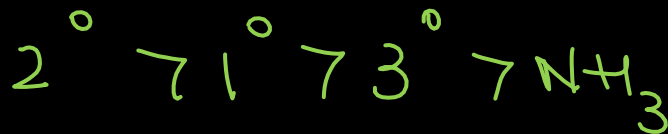
$$\text{Strength of base} \propto \frac{1}{\text{-I power of group present on } -\text{NH}_2}$$
$$\propto K_b$$

Basicity of Amines in aqueous Phase

To determine the basic strength of amines in aqueous phase we have to consider **inductive effect**, **solvation effect** and **steric hindrance**. The order of basic strength is therefore experimental in aqueous state as we can't give priority to stability provided by any one factor. Two results are important for aqueous phase :



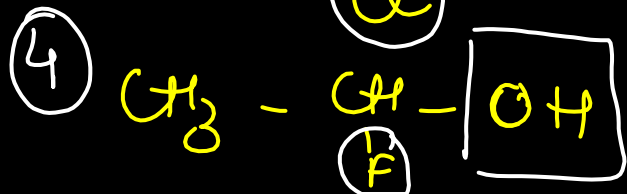
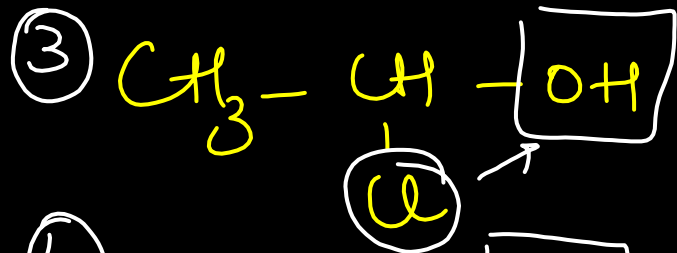
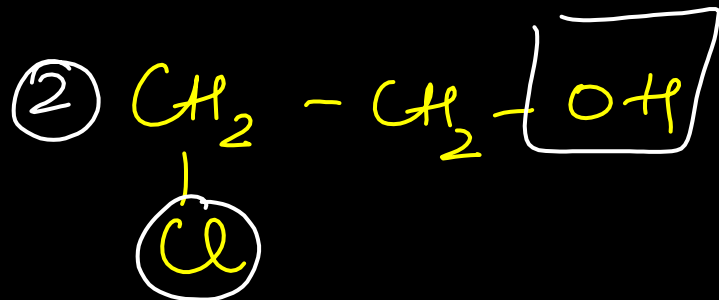
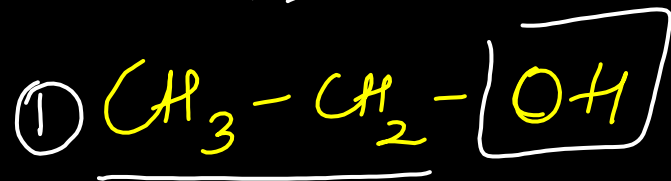
Polar solvent



Remember

SUBSCRIBE

+I



Acidic strength



ABHIMANYU BATCH-2026 SELECTION TEST *IS HERE!*



Choose your slot & secure your chance
to join the most awaited batch of the year



SELECTION TEST 1
16TH AUGUST



7:00 PM
ONWARDS



SELECTION TEST 2
23RD AUGUST



11:00 AM
ONWARDS

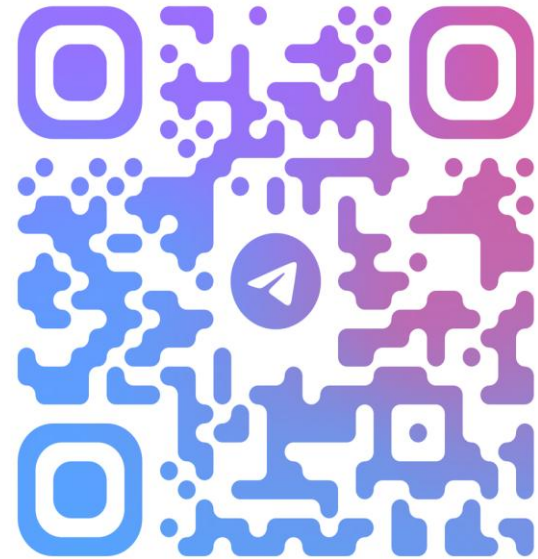


JOIN THE **VEDANTU SCIENCE OLYMPIAD**
TELEGRAM CHANNEL
TO GET THE TEST LINKS, UPDATES,
AND IMPORTANT INSTRUCTIONS.



TELEGRAM LINK IS PINNED IN THE COMMENTS

DON'T MISS THIS OPPORTUNITY TO BE A PART OF
ABHIMANYU BATCH. SEE YOU IN THE TEST! 



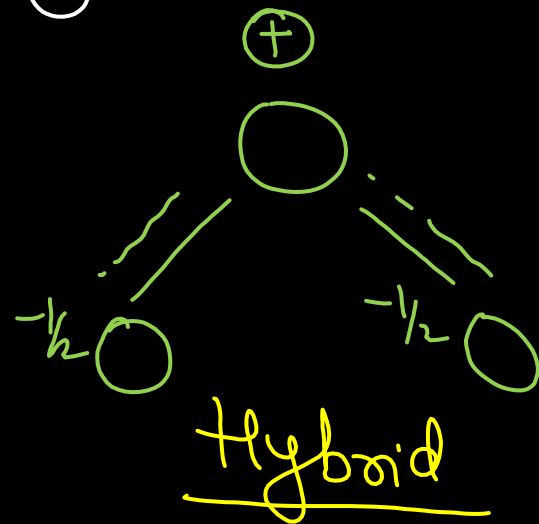
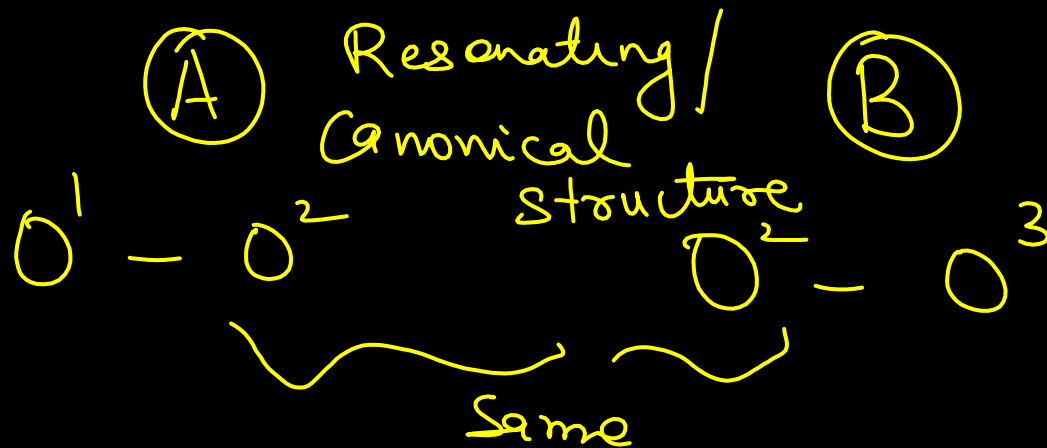
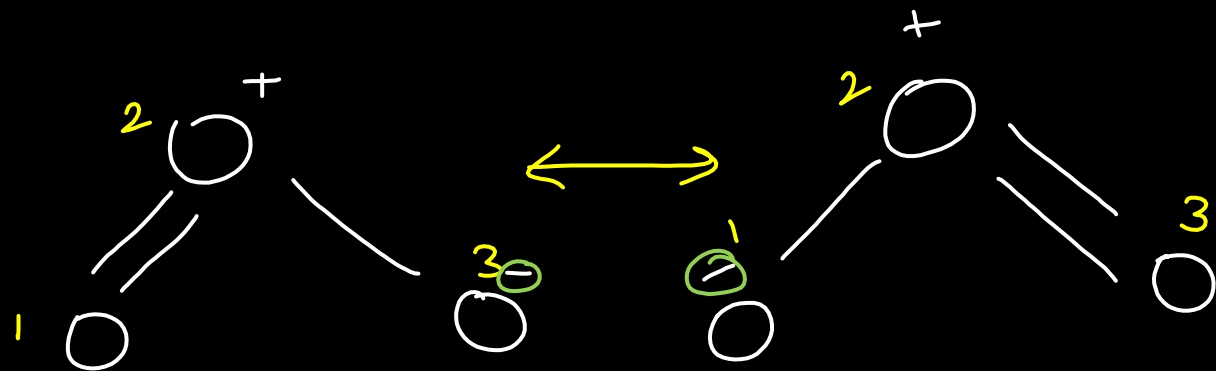
@VEDANTU_SCIENCE_OLYMPI
AD

Resonance

Most Important

SUBSCRIBE

Ozone O_3



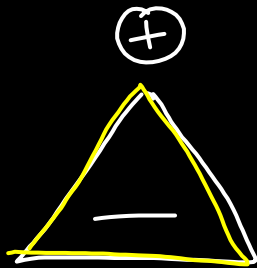
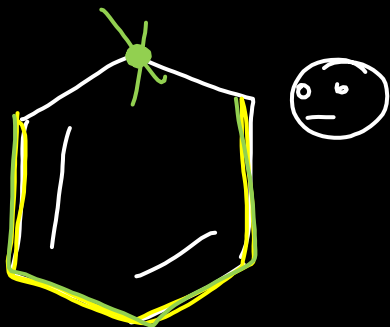
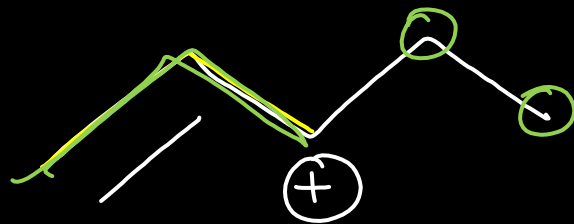
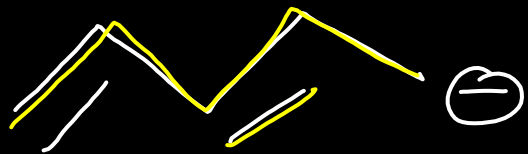
RESONANCE

↓
What → delocalisation of charge

↓
Why → to attain stability

↓
Where → conjugated system

↓
How

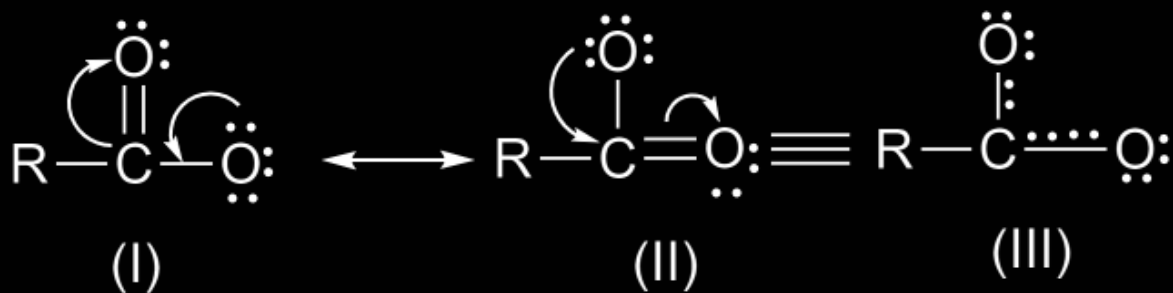


atom to participate
in resonance →

pie	l. p
⊖	odde-
⊕	

Resonance:

When two or more contributing structures of compound can be drawn such that none of the structure alone can explain all its properties then we draw an average structure of all contributing structures called as “Resonance Hybrid” and Contributing structure are called as “Resonating structures” OR “canonical structures” and the phenomenon is called as “Resonance”



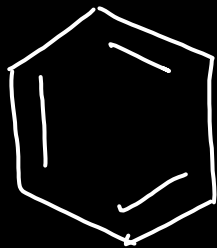
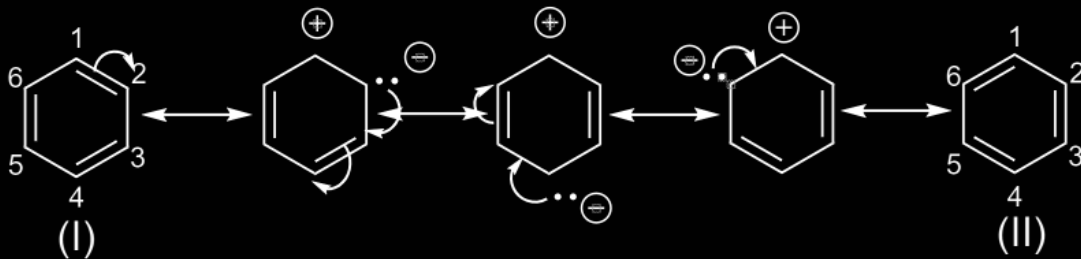
Resonating
Structures

Resonance hybrid

SUBSCRIBE

Conditions for Resonance

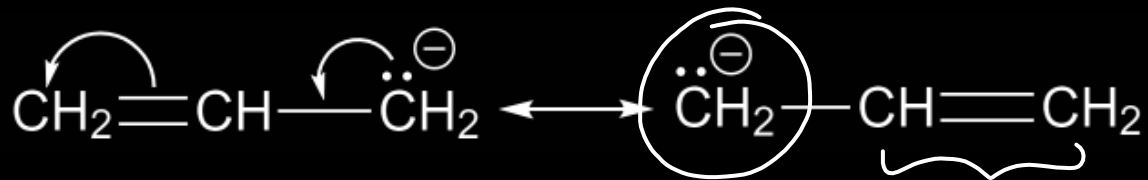
👉 π alternate π bond :



Conditions for Resonance

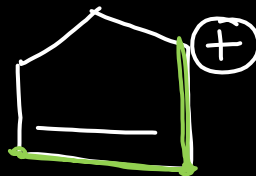
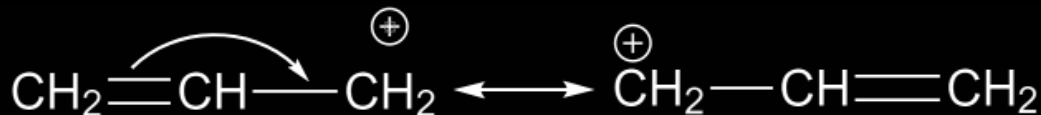
👉 π bond alternate negative charge :

Example : Allyl carbanion



👉 π bond alternate positive charge :

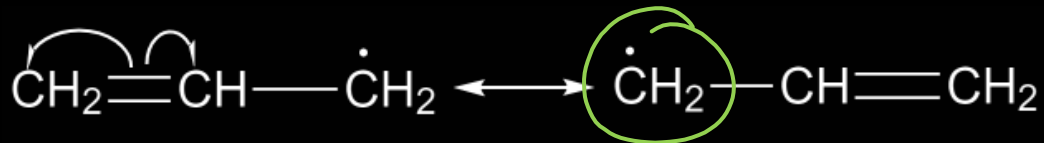
Example : Allyl carbocation



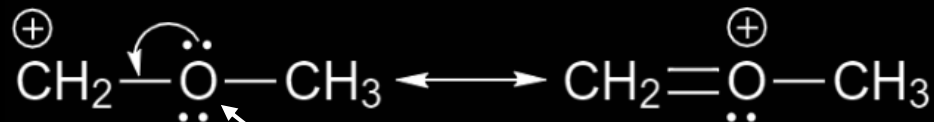
Conditions for Resonance

👉 pi bond alternate odd e⁻

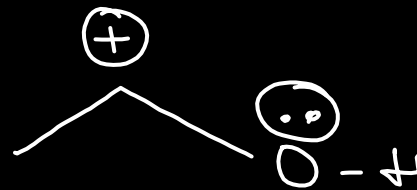
Example : Allyl free radical



👉 Lone pair alternate positive charge



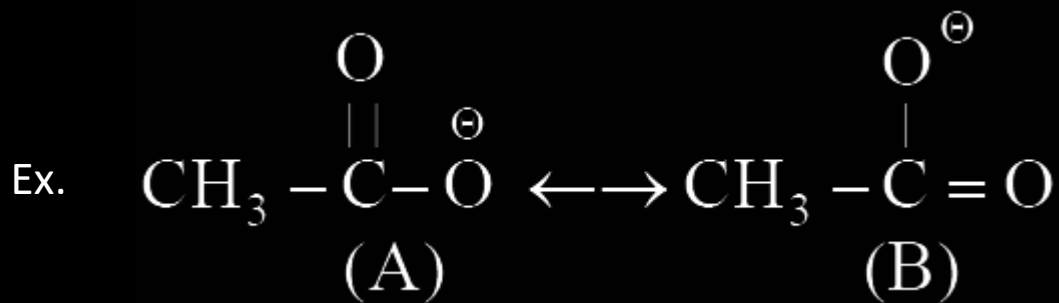
Vacant orbital



SUBSCRIBE

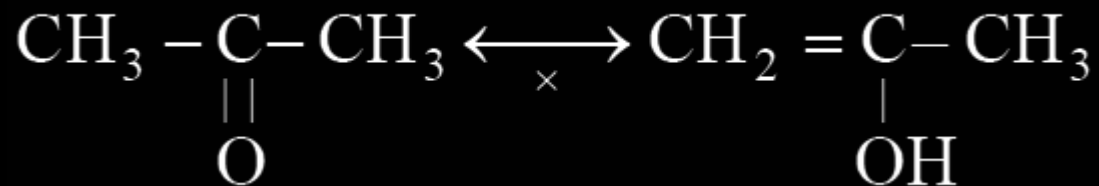
Rules to draw valid resonating structures

- ❖ Electron flow occurs from high electron density to low electron density.
Negative charge $\rightarrow$ Lone pair electron $\rightarrow$ π -bond electron δ^+ positive charge
- ❖ Position of atom cannot be change, only the π -electrons are delocalised.



A & B are Resonance forms.

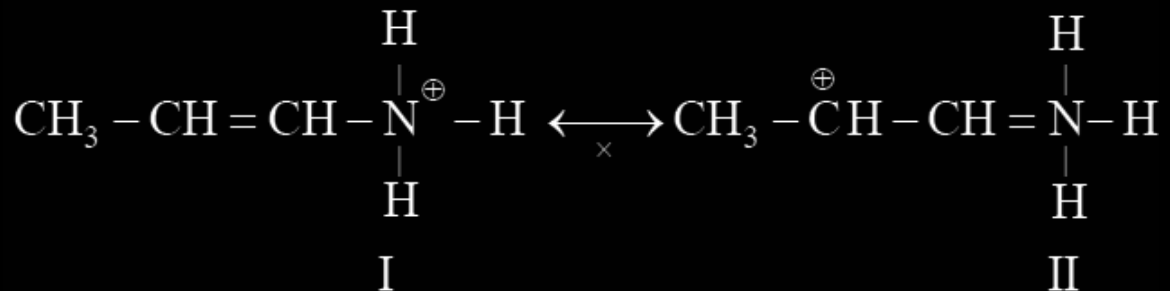
Rules to draw valid resonating structures



Above forms are not resonating forms because position of H is changed.

Rules to draw valid resonating structures

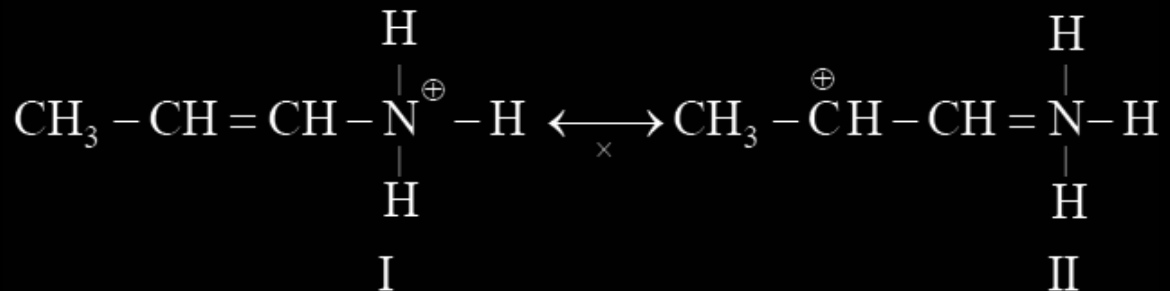
- ❖ Second period elements should not violate the octet rule while drawing resonating structures



Above structures are not resonance form because in structure II, nitrogen has 10 valence electrons which violates octet rule.

Rules to draw valid resonating structures

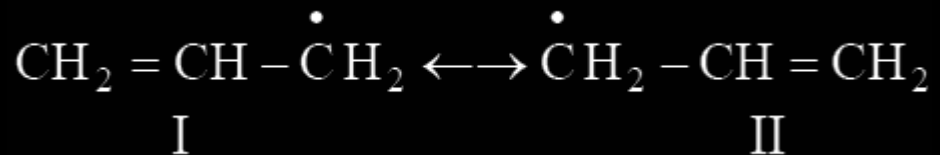
- ❖ Second period elements should not violate the octet rule while drawing resonating structures



Above structures are not resonance form because in structure II, nitrogen has 10 valence electrons which violates octet rule.

Rules to draw valid resonating structures

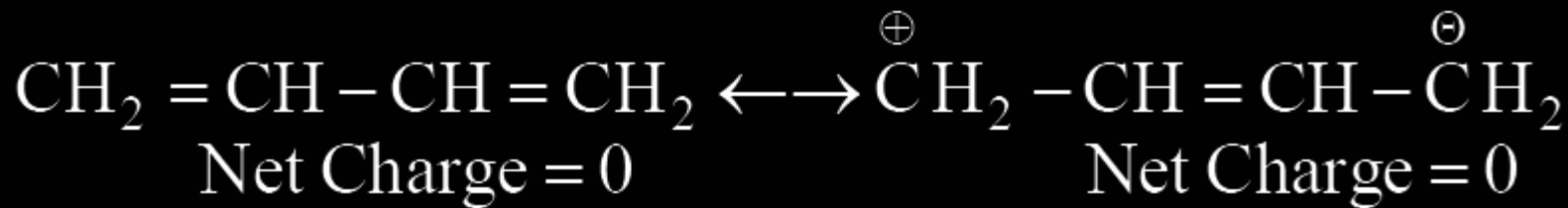
- ❖ Each Resonating structures must have the same number of unpaired and paired electrons.



I & II are resonating structure because both have 8 paired electrons and 1 unpaired electron

Rules to draw valid resonating structures

- ❖ Net charge should be conserved in all resonating structures.



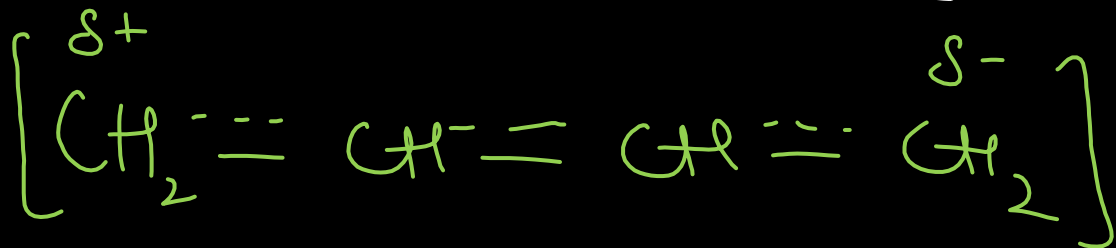
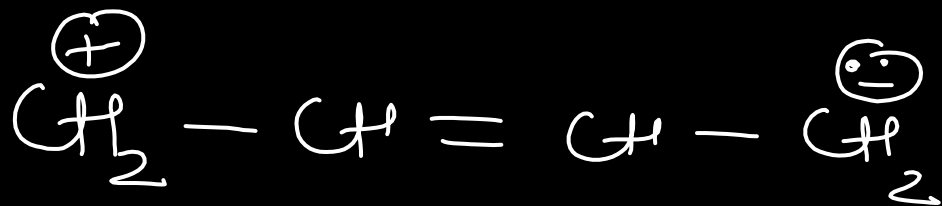
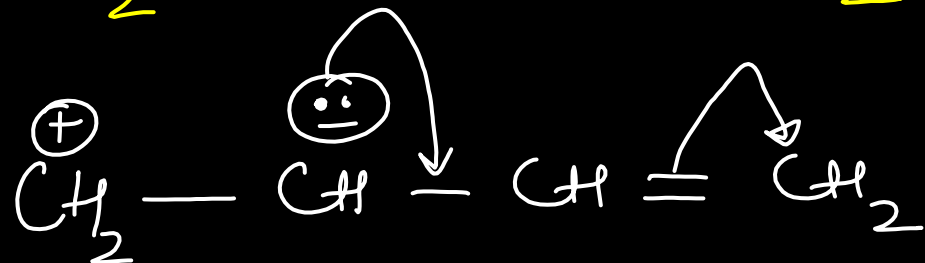
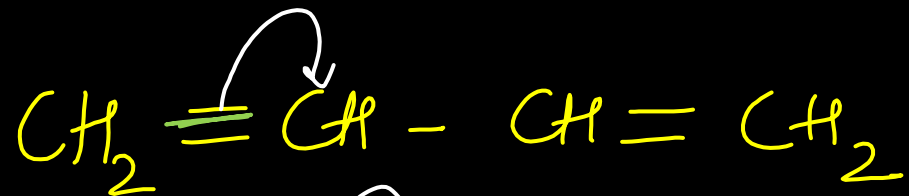
Both structures have same net charge so both are resonance forms.

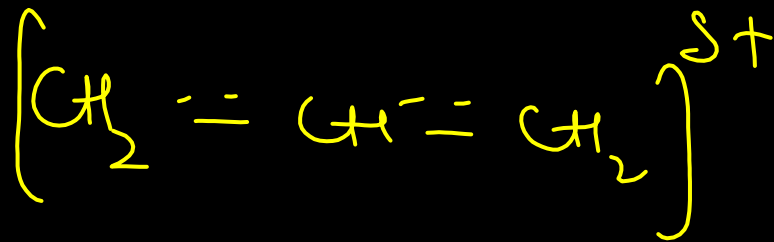
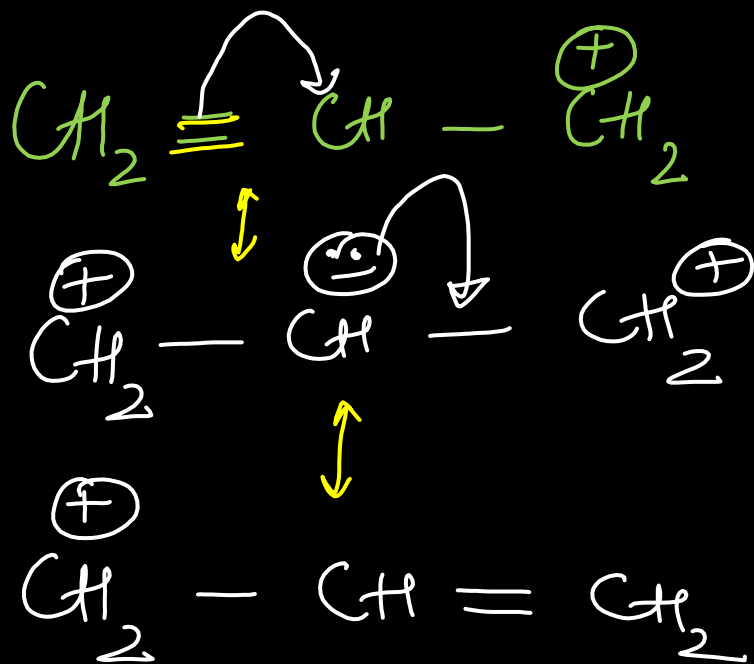
* Remember atoms participating in resonance must be COPLANAR

Hybridisation (sp/sp^2) $\rightarrow$

[SUBSCRIBE](#)

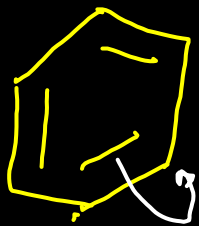






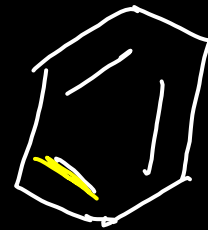
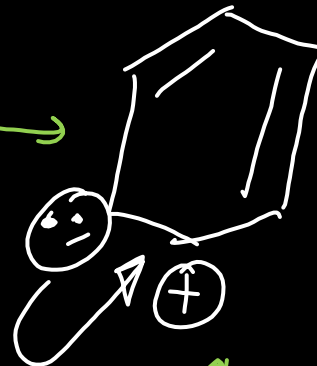
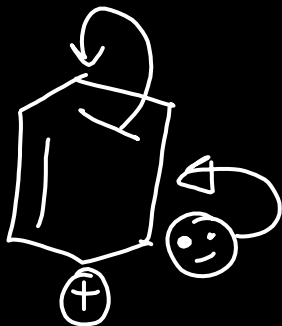
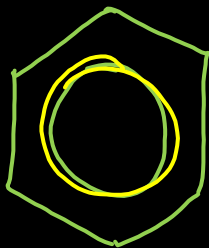
e^- flows from
 high $\rightarrow$ low
 e^- density



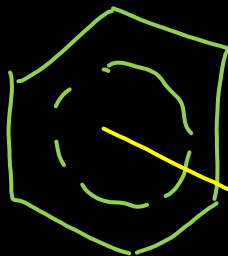


Benzene

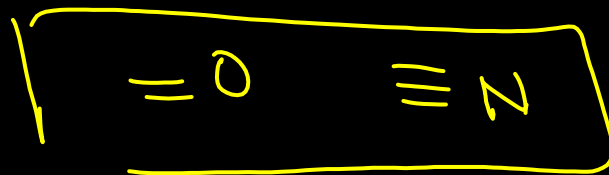
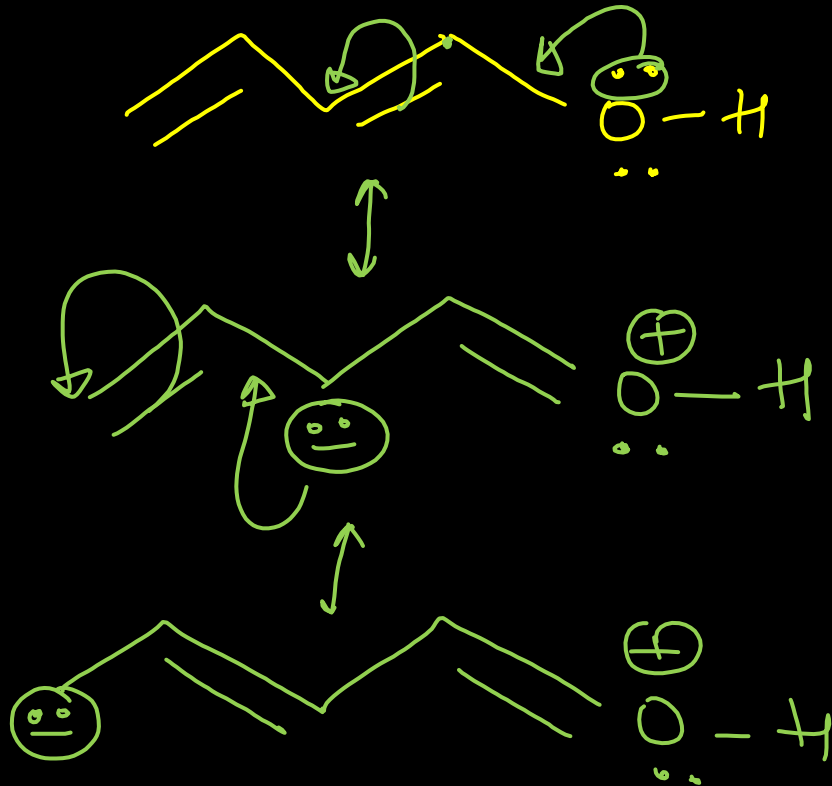
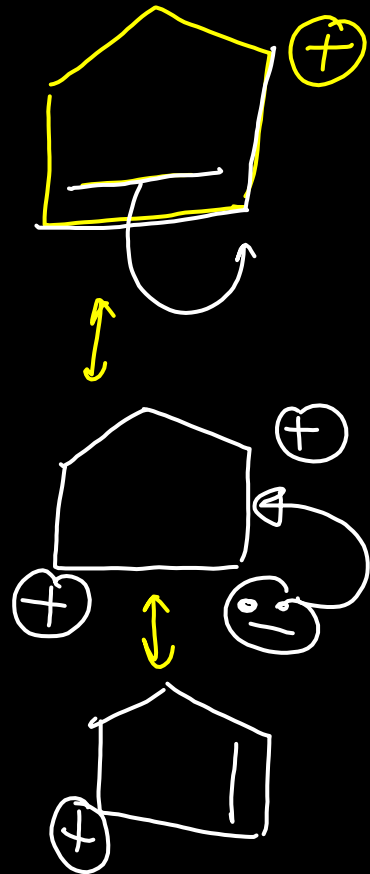
A



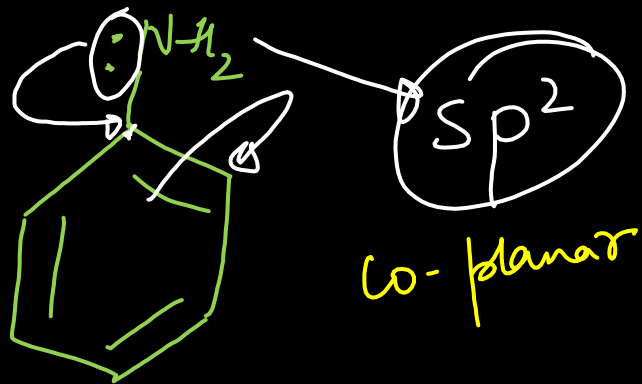
B



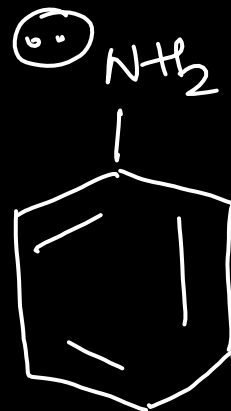
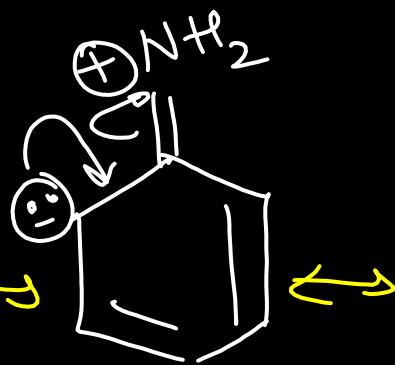
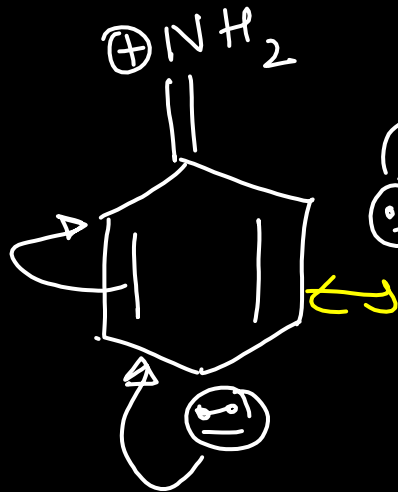
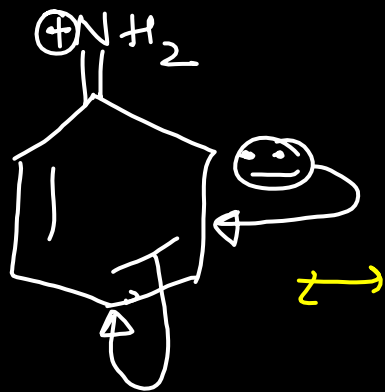
Resonance



e^- flows towards



Aniline



Rules for stability of resonating structures

- more octet complete
 - more no of bonds
 - -ve on more electronegative atom
 - opposite charges → as near as possible
Same charges → as far as possible
- (Follow this in same order)

Rules for stability of resonating structures

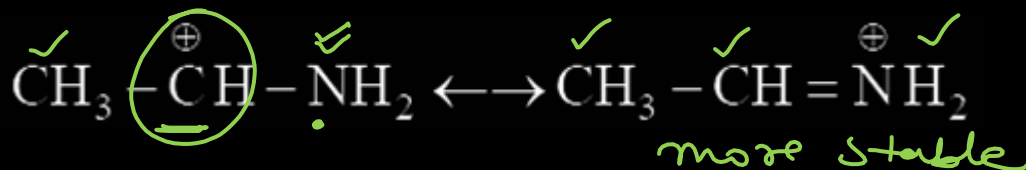
- ❖ Among various resonating structure non- polar resonating structure is more stable than polar resonating structure more number of covalent bonds is considered to be more contributing than others.



more
stable
(major contributor)

Rules for stability of resonating structures

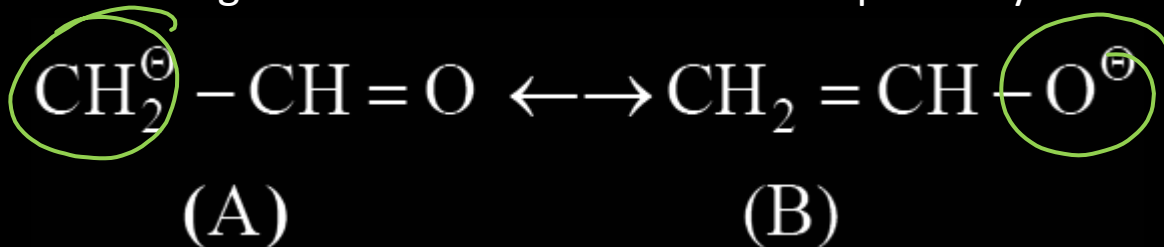
- ❖ If two resonance structures have same number of bonds then that structure in which octet of every atom is complete is more contributing than structure having incomplete octet.



- ❖ If resonating structures have same number of bonds then the neutral structure is more stable than charged structure and if both are charged Resonating structure then less charged resonating structure is more stable.

Rules for stability of resonating structures

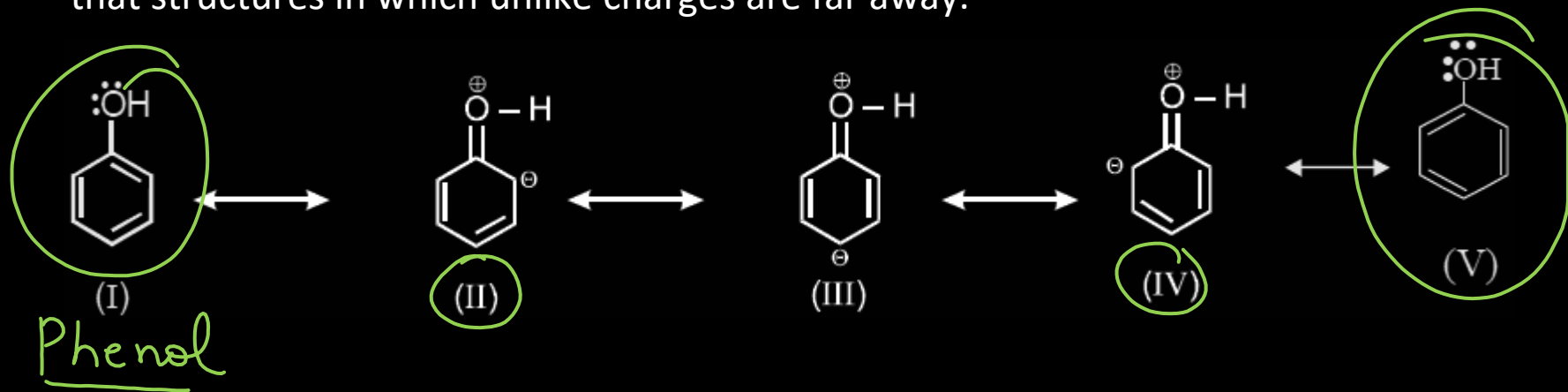
- ❖ In case of charged resonating structures, the most stable one is that in which the positive and negative charge reside on the most electropositive and most electronegative atoms of the structure respectively.



stability of B > A

Rules for stability of resonating structures

- ❖ Resonating structure in which unlike charges are closer is more contributing than that structures in which unlike charges are far away.



Hence stability of II and IV will be the same and both will be more stable than III. The order of stability of resonating structures in decreasing order will be as follows :

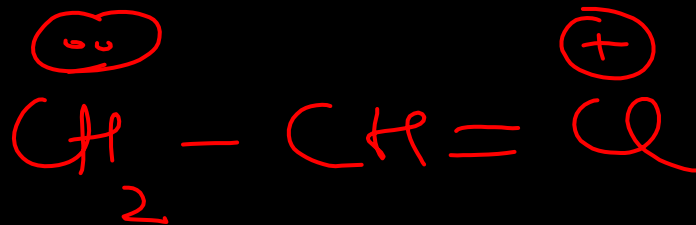
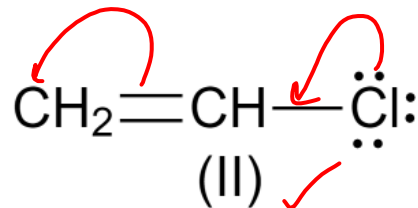
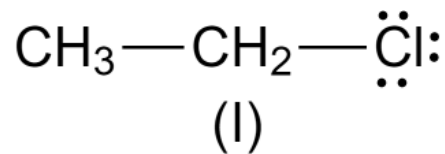
$$I = V > II = IV > III$$

Rules for stability of resonating structures

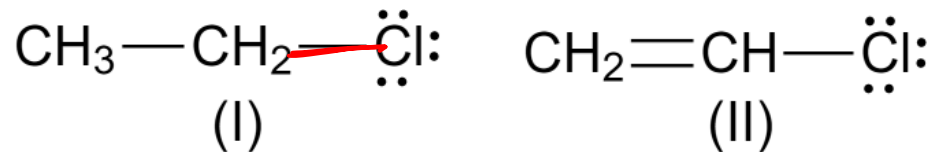
- ❖ Resonating structure in which like charges are farer away is more contributing as compared to structure in which like charges are closer.



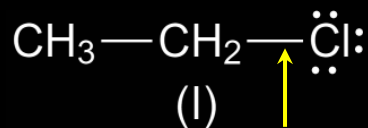
Which of the following structure, has greater C - Cl bond ?



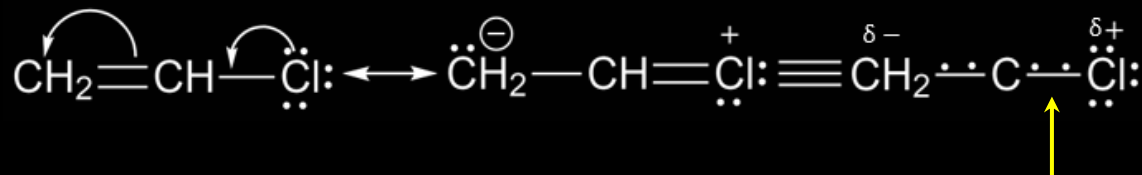
Which of the following structure, has greater C - Cl bond ?



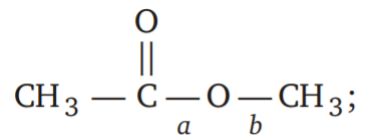
Solution :



Single bond character

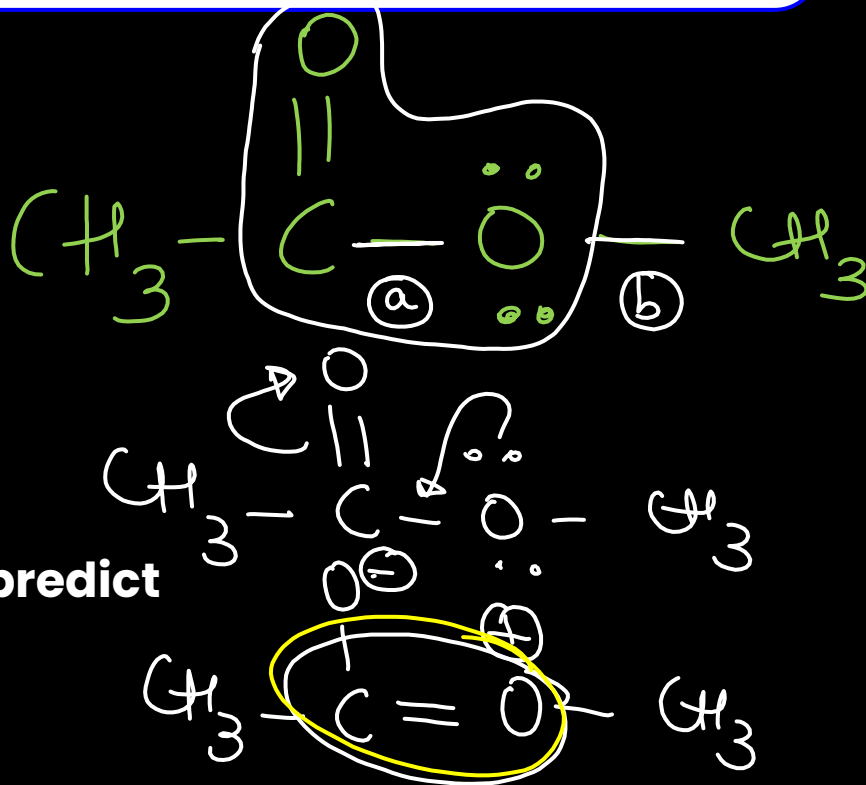


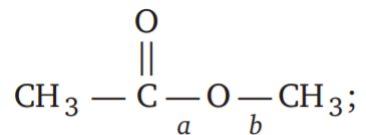
Partial double
bond character



The correct relation between the bond lengths a and b is:

- A** $a = b$
- B** $b > a$
- C** $b < a$
- D** Impossible to predict





The correct relation between the bond lengths a and b is:

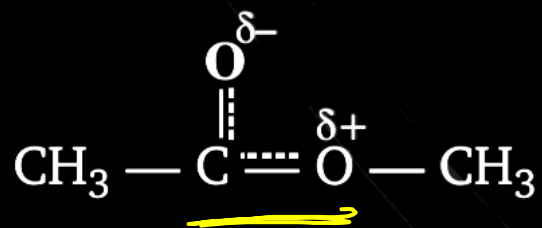
A $a = b$

B $b > a$

C $b < a$

D Impossible to predict

Ans: (b)



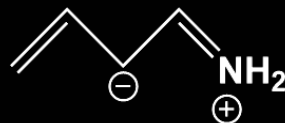
due to resonance this bond has partial double bond characters

Determine the order of stability of the following resonating structure.

(I)



(II)



(III)



A

I>II>III

B

III>II>I

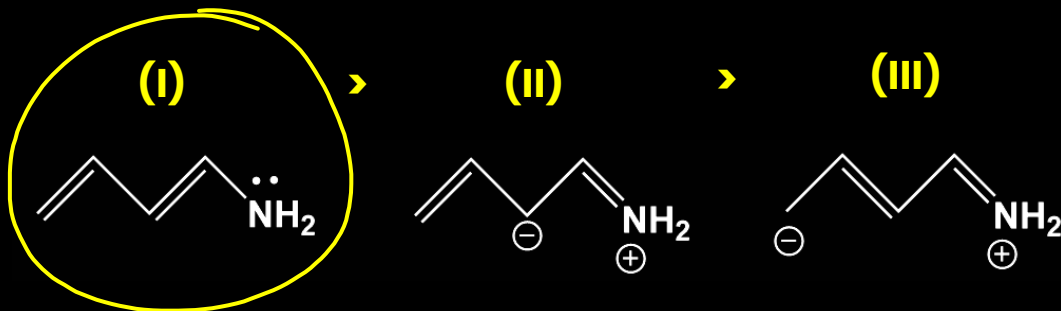
C

I>III>II

D

II>I>III

Determine the order of stability of the following resonating structure.



- Charge separation reduces the stability of resonating structures.
- If charge separation exists, the structure with opp. charges closer is more stable.
- Thus I > II > III

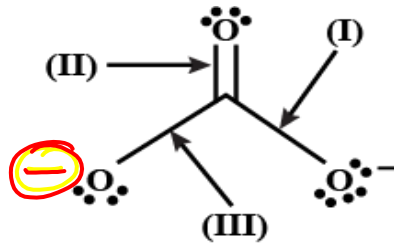
A I > II > III

B III > II > I

C I > III > II

D II > I > III

Taking into account of hybridization and resonance effect, rank the following bonds in order of decreasing bond length.



A

I > II = III

B

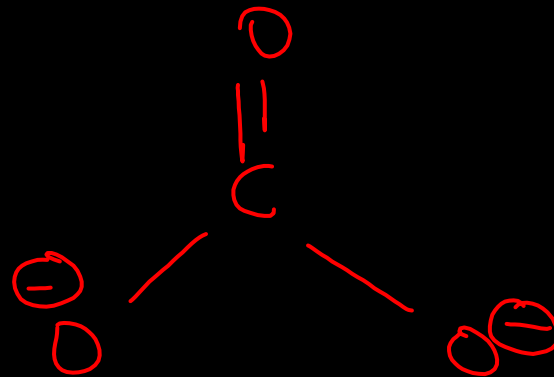
II > III > I

C

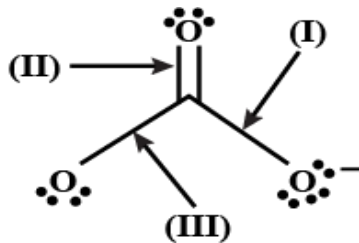
I > III > II

D

II = III = I



Taking into account of hybridization and resonance effect, rank the following bonds in order of decreasing bond length.



A

I > II = III

B

II > III > I

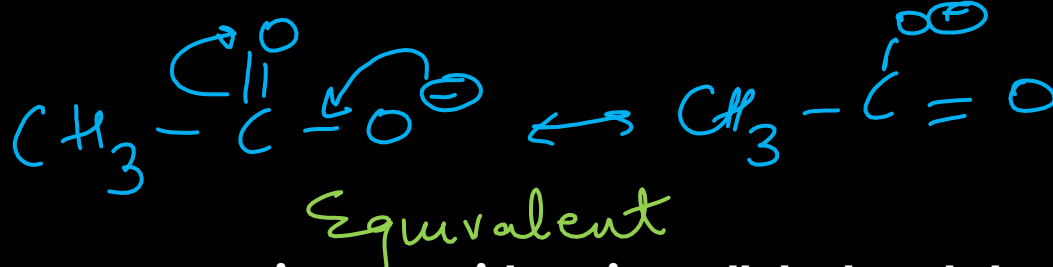
C

I > III > II

D

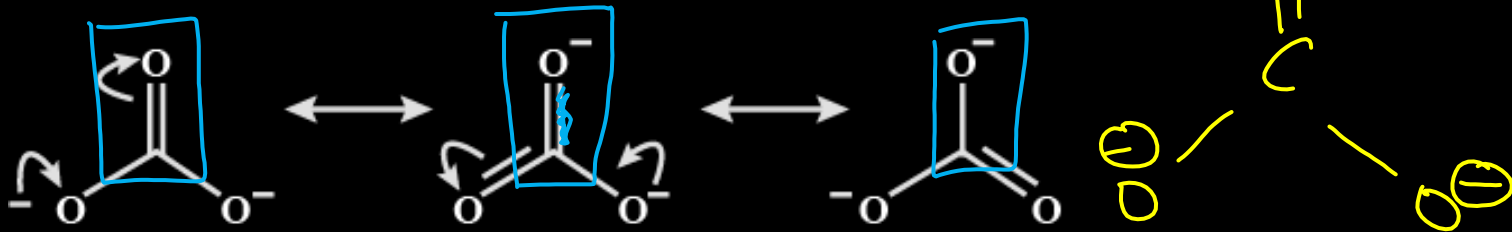
II = III = I

Solution:



Taking resonance into consideration, all the bonds have partial double bond character. As seen by resonance, all the C-O bonds are identical and have the same bond length.

$\therefore \text{II} = \text{III} = \text{I}$ [order of bond length]



Equivalent Structures

Avg.
Bond
Order

$$\frac{2+1+1}{3} = \frac{4}{3} = 1.3$$

The least stable canonical structure among these is:

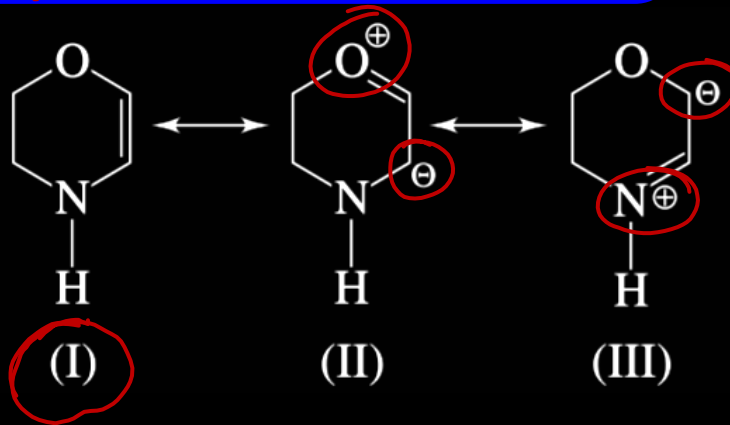
→ minor contributors

A I

B II

C III

D all are equally stable



most stable N

0

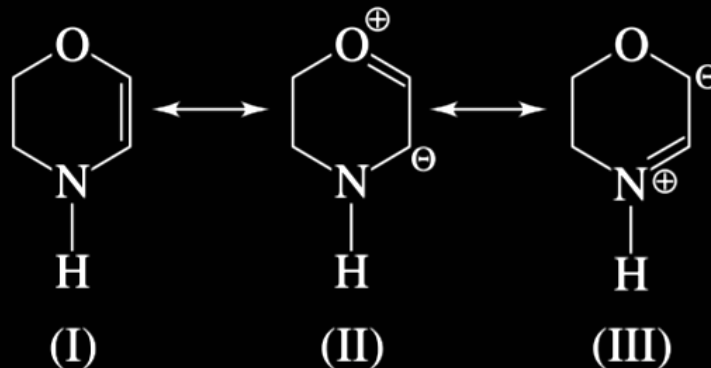
The least stable canonical structure among these is:

A I

B II

C III

D all are equally stable



Ans: In structure II more electronegative element carries a positive charge , hence least stable.

ABHIMANYU BATCH-2026 SELECTION TEST *IS HERE!*



Choose your slot & secure your chance
to join the most awaited batch of the year



SELECTION TEST 1
16TH AUGUST



7:00 PM
ONWARDS



SELECTION TEST 2
23RD AUGUST



11:00 AM
ONWARDS

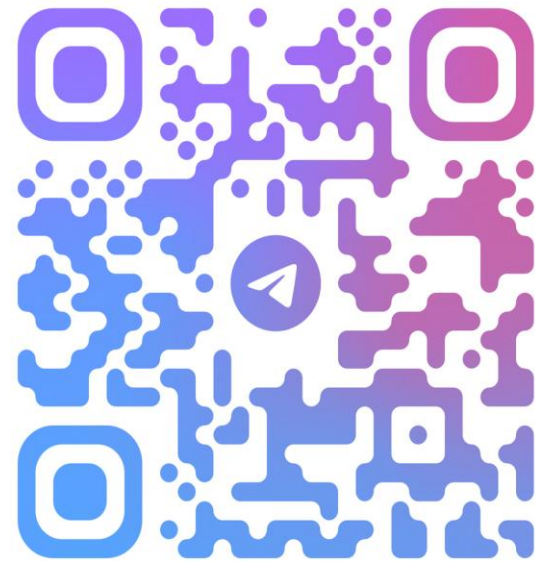


JOIN THE **VEDANTU SCIENCE OLYMPIAD**
TELEGRAM CHANNEL
TO GET THE TEST LINKS, UPDATES,
AND IMPORTANT INSTRUCTIONS.



TELEGRAM LINK IS PINNED IN THE COMMENTS

DON'T MISS THIS OPPORTUNITY TO BE A PART OF
ABHIMANYU BATCH. SEE YOU IN THE TEST!



@VEDANTU_SCIENCE_OLYMPI
AD

 **Don't Miss Out!**



SUBSCRIBED

Hit **LIKE**, **SHARE**, and **SUBSCRIBE**
for more amazing sessions!

